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Leveraging the power of AI

Team Details:

a. DOT CODERS
b. R. DIVAKAR
c. PROBLEM STATEMENT : OPEN INNOVATION

“ Drone Flyability & Risk Analyzer “

Using Google Technologies for Safer Drone Operations

Domain: Maps · AI · Decision Support

Problem Statement & Brief Solution

Problem Statement

- Drone pilots lack an easy way to determine **whether a location is safe to fly**
- Existing regulations are complex and not beginner-friendly
- Accidental violations can cause **safety risks and legal issues**

Our Solution

A **map-based decision-support system** that:

- ❑ Takes **Google Earth coordinates**
- ❑ Analyzes multiple risk factors
- ❑ Classifies regions as **Flyable, Conditional, or No-Fly**
- ❑ Explains the decision in **simple language**

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Opportunities :

Why This Solution is Different

- Most tools only show maps or static no-fly zones
- Our solution provides **risk-based reasoning**, not just rules
- Focuses on **usability for beginners**

How It Solves the Problem

- ❑ Reduces unsafe drone usage
- ❑ Helps users make informed decisions
- ❑ Encourages responsible drone flying

Features Offered :

- Interactive Google Map for location selection
- Real-time coordinate extraction
- Flyability status classification *:*
  - 🟢 Flyable
  - 🟡 Conditional
  - 🔴 No-Fly
- Clear risk explanation for each result
- AI-powered suggestions for safer nearby locations

Google Technologies used in the solution

Google Stack

Google Maps / Google Earth SDK
Location visualization and coordinate selection

Gemini API
Converts rule-based decisions into human-readable explanations

Firebase
Stores predefined risk zones and analysis history

➔ All services used are available under **free tiers**

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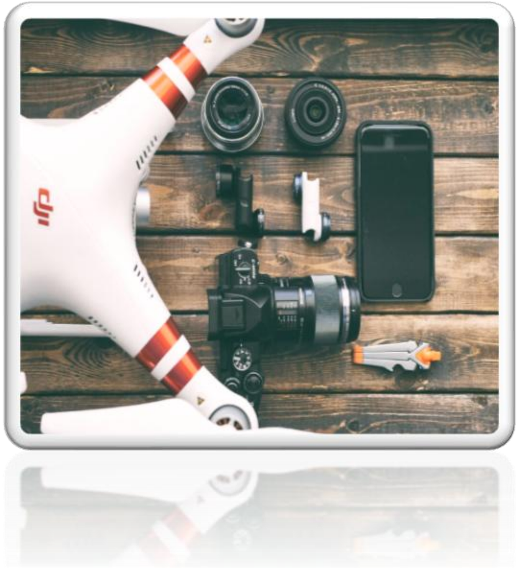
Process flow diagram or Use-case diagram



START

1) User Select Location on Map

2) System extracts Latitude and Longitude

3) Risk Analysis Rules Applied

4) Ai Generates Explanation

5) Flyability Result Displayed to Users

END

Drone Flyability & Risk Analyzer – System Architecture

Processing flyability directly with Google technologies

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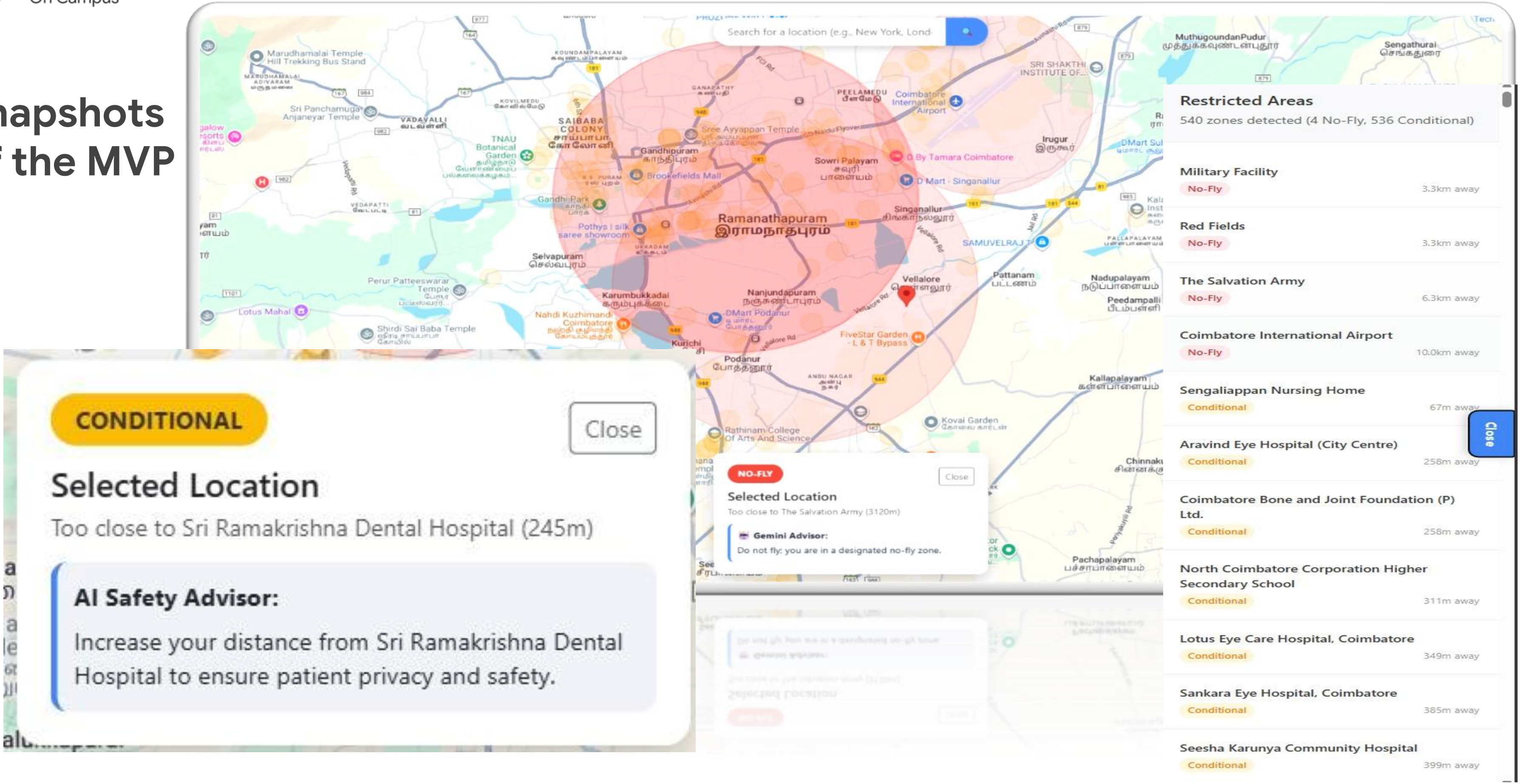
Snapshots of the MVP



Project Links :

1. GitHub Public Repository : https://github.com/Gunavarthan/Have-A-Safe-Flight
2. Demo Video Link : https://youtu.be/LnDv-ubyGfM
3. MVP Link : https://youtu.be/LnDv-ubyGfM

Additional Details/Future Development

- ❑ Integration with live aviation & weather data
- ❑ Altitude-based risk analysis
- ❑ Country-specific drone regulations
- ❑ Mobile application support
- ❑ Offline-first mode for remote areas

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Thank you!